

Cyber Security 1: Exam One

The exam started at 13:02.

Answer by selecting one option in each question. After intermediate saving, you will see your selection. You can make any number of savings. At least one must be done before final saving becomes possible. Do not navigate except through buttons or this link: ([General instructions on its own tab.](#))

In the real and practice exams you can also express uncertainty of your answer to halve the eventual plus or minus points, and tell – for research purposes – whether you remember the question, without effect to points.

1. Encryption is a method for protecting confidentiality of data, and it requires a key, that

- ☐ can even be computed from a human-memorable password if the data is not to be sent anywhere but protected in local storage.
- ☐ has the same length in bits as the data.
- ☐ can be sent along the message as such because the communicating parties must have agreed on such parameters for the algorithm that makes it secret.
- ☐ is different for each message or chunk of data.

2. The basic goal of a public key infrastructure is to

- ☐ bind the public key to the entity to which the key belongs.
- ☐ allow for a reasonably short and easy-to-find chain of certificates.
- ☐ bind a private key to a corresponding public key.
- ☐ enable a wide user base by using a hierarchical structure.

3. Non-repudiation is

- ☐ the term used for cancelling some of a user's rights in an information system.
- ☐ the time shortly before a public key certificate expires and a new one should be ordered.
- ☐ an attack against privacy.
- ☐ a special security goal.

4. An implementation of ____ that contains all the necessary software, protocols, algorithms and keys, is called ____ .

- ☐ a cryptosystem, cryptanalysis
- ☐ cryptanalysis, cryptology
- ☐ cryptology, cryptanalysis
- ☐ cryptography, a cryptosystem

5. If a remote user's machine fails to authenticate the target machine, it may be a wrong one and all of the following are potential threats. Which one is the most serious? The target machine can

- ☐ send malware to the user's machine.
- ☐ be a bot network server.
- ☐ target the user with a DoS attack.
- ☐ give the user's authentication information to an attacker.

Intermediate save

6. What is not a way to protect intellectual property?

- ☐ licensing
- ☐ privacy regulations
- ☐ DRM, digital rights management
- ☐ patents

7. There is a vast variety of things that can be connected to the USB port of a computer. Ignoring problems with electricity, which of the following connections has the least risks in information security?

- ☐ Charging the new camping shower you just bought from a web shop. (This is a little pump with a hose.)
- ☐ Your private memory stick when you read and write document files for long term storage.
- ☐ The optical mouse that your IT staff checked for you to bring home from your office for remote work.
- ☐ Your own mobile phone when you only charge its battery in a locked box at a local supermarket.

8. Malware that does not replicate, pretends to be performing a legitimate action, but does something else in the background is called

- ☐ a Trojan horse.
- ☐ a logic bomb.
- ☐ a virus or a worm.
- ☐ a trapdoor.

9. Password salt is

- ☐ the random number stored without encryption together with each hashed password.
- ☐ the method to make fixed passwords variable by requiring to insert time dependent punctuations.
- ☐ the secret constant that a password checking procedure uses as an input together with the password.
- ☐ the small variation allowed in password systems that use very long passphrases.

10. Which of the following does not apply?

- ☐ An RSA signature cannot be made unless you know the two prime numbers that make up the public modulus n .
- ☐ If you raise a number x smaller than the RSA modulus n to the private RSA exponent (d) and then to the public exponent (e) and only then reduce the result modulo n , you obtain x .
- ☐ The RSA decryption algorithm need not know the two prime numbers that make up the modulus n .
- ☐ RSA encryption can be calculated for a larger number than the public modulus, but decryption still produces a smaller number than the modulus.

11. Man in the middle is an attack type where

- ☐ the attacker or his process relays modified messages between two unknowing communication parties.
- ☐ a process listens to a program's system calls and their responses, and sends the divulged sensitive data to the attacker.
- ☐ a process captures system calls, modifies them, sends them to the OS kernel, and likewise filters the responses to the calling procedure.
- ☐ a cryptographic algorithm is broken at about a square root of effort by working both from the start and end toward the middle.

12. In the cell phone system the network authenticates the phone or actually the SIM card in it. What mechanism is in place here?

- ☐ Challenge-response method that applies a public key system.
- ☐ One-time password
- ☐ Fixed password
- ☐ Challenge-response method, with a symmetric key system.

13. The cryptographic hash value calculated from a message represents the entire message in the sense that

- ☐ a change at any location in the message can be corrected based on the hash.
- ☐ it cannot be obtained from any other message.
- ☐ small changes in the message change the value only slightly.
- ☐ it is very unlikely to find any other message with the same hash.

14. It is important to have an off-site backup copy of files to

- ☐ reduce the possibility of data theft.
- ☐ speed up the process of accessing files at any time.
- ☐ prevent the loss of data in the event of a fire.
- ☐ improve accessibility of files from other locations.

15. It has been said of the password that it should be treated like your own toothbrush. How many of the following password-related features does this statement represent? (i) Entropy, (ii) usability, (iii) keeping it for personal use only, (iv) changing it often enough.

- ☐ one
- ☐ all
- ☐ two
- ☐ none

16. The term attack vector

- ☐ refers to the attacker's point of view to the chain of protections that implement the defence-in-depth approach.
- ☐ refers to the method of an attack - one that happened or is possible.
- ☐ is just a more fancy way of referring to a particular attack that has happened.
- ☐ refers to the combination of all vulnerabilities that exist in a particular information system.

17. DES, the Digital Encryption Standard, was for a long time the state-of-the-art method to

- ☐ carry out the exchange of cryptographic keys.
- ☐ produce confidentiality of communications.
- ☐ authenticate the owner of the secret encryption key.
- ☐ transform digital images to contain a watermark.

18. With the exception of the cookie itself or other cookies, a web browsing cookie cannot break

- ☐ the availability of your files, but it may break their confidentiality or integrity.
 - ☐ the integrity of your files, but it may break their confidentiality.
 - ☐ either the integrity or confidentiality of your files.
 - ☐ the confidentiality of your files, but it may break their integrity.
-

19. Who is responsible if you share in social media something that is legal but not true and this causes harm to someone who reads it?

- ☐ The reader, who took the post for true.
 - ☐ No one.
 - ☐ You, but only if you created the post yourself or reposted something that was not published for the same reader earlier.
 - ☐ You, without conditions.
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20. Privacy is synonymous to

- ☐ secrecy.
 - ☐ privilege.
 - ☐ anonymity.
 - ☐ none of these.
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21. All of the following attacks have some social aspect, but which description gives the best coverage for the concept of social engineering?

- ☐ Inferring sensitive personal or organizational information from discarded papers and media together with public sources.
 - ☐ Getting physically close and even familiar to people in order to see what they input as credentials to information systems.
 - ☐ Exercising the art of influencing people to act against their security policy.
 - ☐ Soliciting sensitive personal or organizational information by persuasive or masquerading emails.
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22. TLS is a very common cryptographic protocol. The T in its name comes from the word

- ☐ transmission.
 - ☐ transport.
 - ☐ trusted.
 - ☐ total.
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23. How does a packet filter react to TCP and UDP port numbers?

- ☐ It passes through the packet if and only if those numbers are equal.
- ☐ It doesn't take them into account.
- ☐ It returns the packet if either number is invalid.
- ☐ It rejects the packet if either number is invalid.

Intermediate save

24. Information assurance is sometimes considered a wider term than information security. On the other hand, *assurance* is just part of information security, namely

- ☐ a synonym for accountability.
 - ☐ a synonym for authentication.
 - ☐ evidence that security mechanisms are efficient.
 - ☐ the level up to which risk management has been able to *transfer* information security risks.
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25. Sandboxing is a security mechanism, where

- ☐ new untrusted software is run in a disconnected test system to check it for malicious behaviour.
 - ☐ untrusted code, which is not signed, is continuously restricted from accessing system resources.
 - ☐ a new version of software is tested by scrutinizing all memory accesses it does that the earlier version did not do.
 - ☐ a new version of software is installed into a production system but is kept from doing anything that the earlier version did not do.
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26. What is true about fires?

- ☐ Fire detection and suppression must be taken care of by the ordinary facilities management and there are no special demands from the information technology point of view.
 - ☐ It is not so much the computers that catch fire, but the batteries and their charging are causing risks especially in mobile devices.
 - ☐ The risk of fire is the foremost reason why backups must be made.
 - ☐ The increase in processing power of computers and in their memory sizes has come with the elevated risk of overheating and fire.
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27. If you need to browse the internet anonymously,

- ☐ you must be cautious because most services for that purpose are against the law, at least in Finland.
 - ☐ you have no choice but do that on a public machine - in a library for instance.
 - ☐ you can simply set your browser to the privacy preserving mode.
 - ☐ the TOR network is an option you can seriously consider.
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28. How can an attack be passive?

- ☐ The attack happens from a compromised slave machine owned by an unknowing user.
 - ☐ The attacker just puts up a web-site asking for donations and starts receiving money.
 - ☐ Actually it cannot be. An attack is by definition active.
 - ☐ A passive attack is such that does not interfere with the integrity or availability of the victim's computing.
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29. Which of the following does not need to be considered in the risk analysis?

- ☐ How much is invested in information security in competing companies.
- ☐ How difficult or expensive it would be for an attacker to carry out certain types of threats.
- ☐ Company's data resources and their value.
- ☐ Impact of security breaches on the company's reputation or public image.

Intermediate save

30. Which counts as two factor authentication?

- ☐ A user name and a PIN.
 - ☐ A hard token and a smart card.
 - ☐ A PIN and a hard token.
 - ☐ A password and a PIN.
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31. GDPR defines several roles with respect to a person's personal data, but not the role of data

- ☐ controller.
 - ☐ subject.
 - ☐ processor.
 - ☐ object.
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32. Steganography is

- ☐ a special branch of cryptographic hash functions.
- ☐ an obsolete method of protecting secret messages.
- ☐ synonymous to digital watermarking.
- ☐ an art of embedding data into a larger quantity of other data.

Intermediate save

Make at least one intermediate saving. After that the ending button for this stage appears here.